Windborne: Fostering Nature Connection and Playfulness through Multimodal Interaction in Healthcare Settings

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Abstract

Hospital environments often involve stressful experiences for patients and visitors. To address this, we developed *Windborne*, a gesture-based multimodal installation that translates invisible wind phenomena into interactive experiences of light and bubbles. Through the development of *Windborne*, we discuss caring design as a framework that shifts the focus from "fixing" patient conditions to "caring" for their immediate emotional and relational well-being within clinical spaces. Grounded in a feminist ethics of care and inspired by the dispersal of dandelion seeds, the system balances ambiguity and feedback and is designed to support open-ended, playful moments for individuals with diverse physical abilities. In this demonstration, we present the implementation of *Windborne*, showcasing how it responds to gestures and movement to trigger light patterns and bubble blow, and explore it as a poetic medium with the potential to support emotional well-being and positive distraction in healthcare settings.

CCS Concepts

• **Human-centered computing** → **Interface design prototyping**; *Interactive systems and tools.*

Keywords

Multimodal Interaction; Caring Design; Playfulness; Bubbles

ACM Reference Format:
Borui Wang and Keith Evan Green. 2026. *Windborne*: Fostering Nature Connection and Playfulness through Multimodal Interaction in Healthcare Settings. In *INTERNATIONAL CONFERENCE ON MULTIMODAL INTERACTION (ICMI '26), October 05–09, 2026, Napoli, Italy.* ACM, New York, NY, USA, 3 pages. https://doi.org/10.1145/3776574.3832531

1 Introduction

Hospital inpatients and visitors often experience a profound state of stress and disconnection that compromises their well-being [9]. Separated from natural environments within the highly constrained context of clinical waiting areas and hospital wards, they face a form of sensory deprivation, losing the restorative effects of a gentle breeze, sunlight, and living plants. Medical routines and physical constraints can erode their sense of comfort, leaving them in passive roles of waiting or receiving treatment. Extended confinement weakens social connections, leaving fundamental desires for lighthearted interaction unfulfilled.

Figure 1: Composite concept of *Windborne* in action

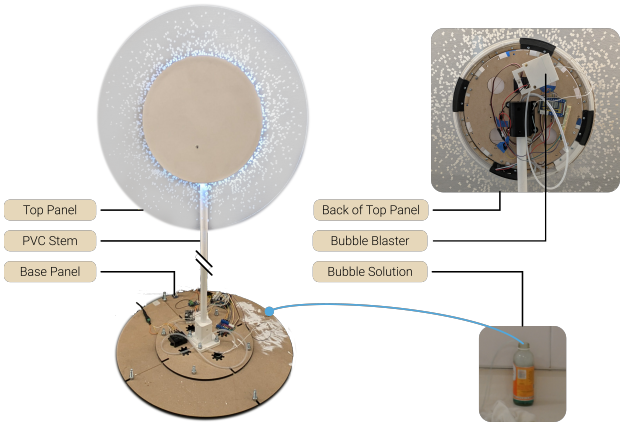

Figure 2: *Windborne*, the interactive installation (PVC stem shortened and does not represent its actual length).

To address these challenges, we introduce *Windborne* (Figure 1), a free-standing multimodal installation designed to support nature connection and playful interaction. We ground this work in a caring design perspective, which deliberately shifts the design paradigm in healthcare settings from "fixing" the human body to "caring" for patients' subjective and relational experiences [5]. Thinking about interactive technology through Tronto's feminist care ethics [8], we view the installation not as an intervention designed to cure medical conditions, but as a poetic medium that attends to fragile relationships and emotional well-being. By translating an invisible breeze into tangible, multimodal experiences through bubbles,

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ICMI '26, Napoli, Italy
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ACM ISBN 979-8-4007-2318-6/2026/10
https://doi.org/10.1145/3776574.3832531

dynamic light, and gesture-based interaction, *Windborne* demonstrates how design can prioritize a practice of care—attending to connections with natural phenomena, a sense of presence, and social interaction—rather than trying to fix or cure.

2 Related Work

We position *Windborne* at the intersection of bubble-based interactions and nature-inspired multimodal interfaces.

2.1 Bubble-Based Interactions in HCI

Soap bubbles have previously been used in HCI research. Their ephemeral, playful, and dynamic nature makes them an appealing medium. Sylvester et al. [7] filled bubbles with smoke and used them as clickable buttons, provoking thinking on control and materiality. *BubbleTalk* [3] explored the ephemeral aspects of human communication through an open-source toolkit that produces bubbles based on spoken words. While these projects explore the fleeting qualities of bubbles, we focus on capturing their lightness to create an engaging experience.

2.2 Multimodal Interfaces Inspired by Natural Phenomena

Prior work has explored nature-inspired elements to promote mindfulness. *Wwall* tracked water consumption to provoke collective awareness, while *The Crock of Shh* [1] leveraged a slow-technology water nozzle coupled with acoustic transducers to promote positive reflection through whispered words. Similarly, light sculptures like *Luminous Wind* [4] and *Shape of Wind* mapped airflow patterns onto macro-scale lighting installations. However, these either operate as passive aesthetic barometers or rely on explicit instruction. Witaya Junma's *Into the Wind* [6] pairs breath inputs with real-time bubble blowing. *Windborne* differentiates itself by substituting direct blowing with gentle, inclusive hand gestures, accommodating individuals with strict physical limitations or attached medical equipment. In contrast to these systems that focus on reflection or mindfulness, *Windborne* is designed to offer dynamic, playful interaction, with the potential of providing sensory relief in clinical environments.

3 System Implementation

Windborne is implemented as a free-standing, human-scale installation (Figure 2) divided into four modules:

Sensing Module. This module captures gestures and presence. Active hand motions are registered via a Grove PAJ7620U2 gesture sensor positioned at the center of the top panel. To detect passive movement, two Grove ultrasonic rangers are embedded in the base panel. These rangers track spatial proximity, allowing the system to monitor the presence and estimate the movement direction of nearby passersby.

Visualization Module. Designed to represent the lightness and delicacy of a dandelion seed head, this module utilizes a transparent acrylic disk edge-lit by a high-density WS2812B LED strip (144 LEDs/meter, approximately 1 m / 3.3 ft long) to achieve uniform illumination [10]. The panel features semi-transparent laser engravings that act as light guide pixels, reflecting light outward, similar to those in LCD backlight panels. To generate these patterns, we built upon the open-source light guide plate algorithm by Vasquez [10], which creates density dot patterns using spline functions. We extended the algorithm with a polar mathematical mapper that translates standard Cartesian XY-plane square spline density matrices into circular coordinates. The prototype supports three panel configurations: *Dot* (organic circular pixels for a friendly feel), *Dash* (concentric arcs of varying lengths designed to evoke the organic swaying of dandelion fluff in a gentle breeze), and *Patch* (overlapping rectangular matrices that mimic the gaps between a dandelion head's radial seeds).

Ephemeral Medium Module. Mounted behind the top panel, this module consists of a Bambu Lab electric bubble blaster kit housed in a 3D-printed casing and mounted on an MG90 servo motor. To produce durable, slow-drifting bubbles, we utilized a custom bubble solution recipe combining dish soap, xanthan gum, corn syrup, and water [2]. We selected this formulation because informal tests demonstrated that it produces bubbles with a longer lifespan than standard commercial alternatives while relying entirely on non-toxic, household ingredients. This solution is stored in a reservoir behind the bottom plate of the base.

Control and Structure Architecture. The core control system includes an Arduino UNO equipped with a Grove connector shield, coordinating the gesture sensor, ultrasonic rangers, the servo motor, and a Grove MOSFET that actuates the bubble blaster. Power is split into two channels: the high-current demands of the bubble blaster and LED strip are met by a dedicated 5V 15A power converter, while the Arduino and sensors are powered separately via an MB102 breadboard power supply to isolate electrical noise and ensure safety. Structurally, the installation connects the top panel and base via a central hollow PVC tube, housing all internal cabling and rubber tubes. This PVC stem provides structural stability while retaining some compliance, allowing the torque generated by the bubble blaster's servo to cause the top panel to sway organically. Critical load-bearing joints are secured with screws and nuts, while components requiring maintenance are secured with Velcro for easy disassembly.

4 Multimodal Interaction Scenarios

To accommodate varying patient energy levels, physical capabilities (e.g., children in wheelchairs or with attached IV lines), and comfort levels, *Windborne* is designed around three interaction scenarios (Figure 3) that provide multi-sensory feedback with minimal physical requirements:

Lateral Hand Waves. When a user stands in front of the installation and performs a slow left-to-right or right-to-left hand wave, the Grove gesture sensor registers the motion. The Arduino then triggers a corresponding light pattern across the WS2812B LED strip, while the MG90 servo motor pans the bubble blaster to release a stream of bubbles along the trajectory of the user's hand. This scenario is designed to evoke the metaphor of dandelion seeds dispersing in a gentle breeze. As the blaster operates, its torque causes the flexible PVC stem to sway gently. (Figure 3a)

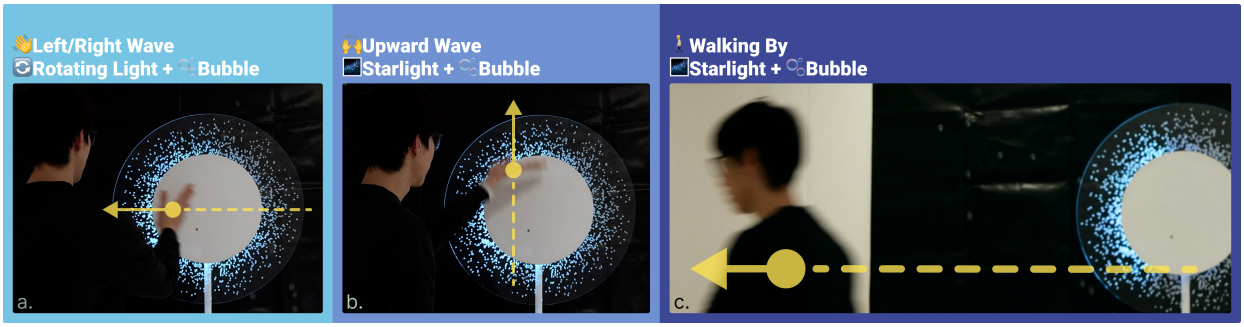

Figure 3: The three main interaction scenarios of the *Windborne* system: (Left) active lateral wave gestures triggering horizontal bubble streams, (Center) active vertical gestures evoking sweeping light and vertical bubble releases, and (Right) passive presence tracking that leaves a gentle light and bubble behind the user.

Vertical Hand Waves. An upward sweeping hand gesture triggers a more dynamic response, emulating a sudden gust of wind. Rapidly blinking "starlight" pixel effects ignite from the base of the circular acrylic panel, sweep symmetrically along the circular edge, converge at the top, and shrink upward. Simultaneously, the bubble blaster performs an extended vertical release of bubbles. (Figure 3b)

Passive Engagement. Recognizing that some patients in waiting spaces may experience too much fatigue or shyness to interact actively, the system passively monitors nearby movement. When someone walks past, the installation detects their direction of motion and generates a soft trailing starlight trail that tracks their movement, accompanied by a brief, synchronized release of bubbles swirling into their physical wake. This is designed to support a low-pressure atmosphere for quiet engagement. (Figure 3c)

5 Preliminary Feedback

We gathered feedback on the *Windborne* prototype through two stages: an online survey ($n = 30$) after watching a video of the installation, and three in-person feedback sessions. In the survey, participants rated overall visual appeal at a mean of 4.1/5.0, curiosity provocation at 3.6/5.0, and potential as a conversation starter at 3.8/5.0 (on a 5-point scale, where higher scores indicate more positive ratings). Over 50% of participants preferred the *Windborne* experience to traditional passive nature videos. Qualitative coding of open-ended responses showed that 21 participants highlighted potential "mood improvement" and 14 identified "distraction" as key benefits. In the in-person feedback sessions, participants reported positive feelings (e.g., "happy", "peaceful", "calm", "relaxed"), noting that the interaction evoked "playful childhood" memories.

6 Safe and Responsible Innovation Statement

As a gesture-based bubble-blowing installation designed for healthcare environments, *Windborne* prioritizes patient safety, privacy, and inclusivity. The system relies entirely on local processing sensors rather than cameras, ensuring that no personally identifiable biometric data are captured or stored, thereby safeguarding patient privacy in sensitive clinical environments. To ensure physical safety, the bubble solution utilizes a non-toxic formula, and the installation is designed to prevent tipping. Additionally, the bottom panel is enclosed to shield participants from exposed screws and circuitry. The system accommodates users of diverse motor and cognitive abilities through low-effort interaction modes. However, deployment requires regular maintenance to prevent slipping hazards from bubble residue.

7 Conclusion

This work presents *Windborne*, a gesture-based multimodal installation that translates invisible wind phenomena into interactive experiences of light and bubbles. With multi-sensory feedback and accommodating diverse physical abilities, the system provides open-ended, playful stimulation. *Windborne* demonstrates a novel direction for enhancing well-being through a playful and caring design approach.

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